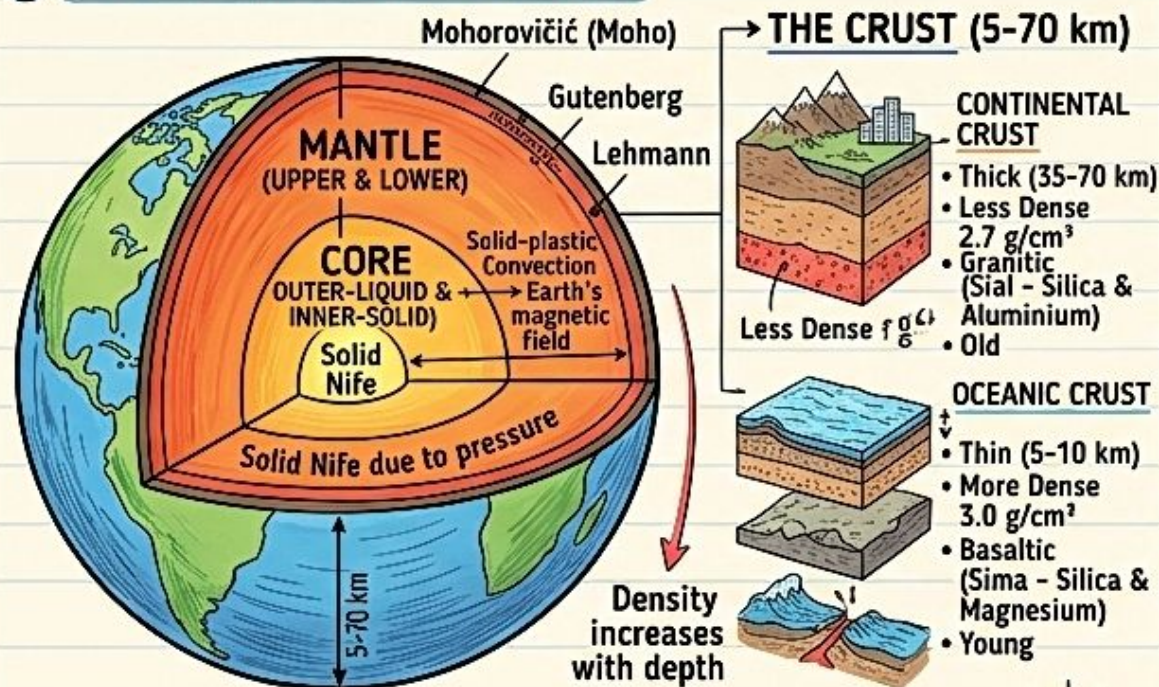


COMPREHENSIVE NOTEBOOK: STRUCTURE OF THE EARTH

ABHISHA AI
by Abhishek Pathak

1 INTERIOR OF THE EARTH



1 THE CRUST - Liquid Nife (Fe & Ni) → Earth's magnetic field



CONTINENTAL CRUST

- Thick (35-70 km)
- Less Dense (2.7 g/cm^3)
- Granitic (Sial - Silica & Aluminium)
- Old



OCEANIC CRUST

- Thin (5-10 km)
- More Dense (3.0 g/cm^3)
- Basaltic (Sima - Silica & Magnesium)
- Young

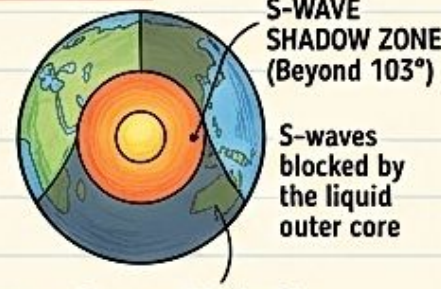
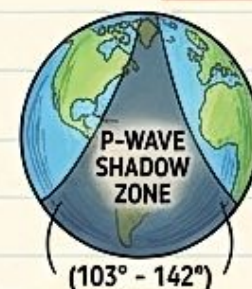
2 & 3. SEISMIC WAVES & SHADOW ZONES

SEISMIC WAVES: X-rays of Earth's Interior

P & S-waves travel through, Surface waves travel along surface



SHADOW ZONES



4. ROCK TYPES & ROCK CYCLE

IGNEOUS ROCKS

- Formed from magma cooling (e.g., Granite, Basalt)



SEDIMENTARY ROCKS

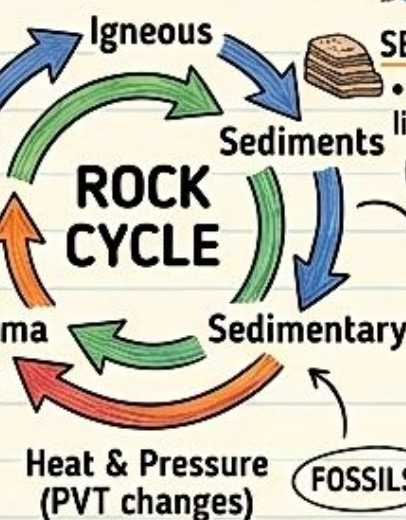
- Formed from lithification of sediments (e.g., Sandstone, Sandstone, Limestone, Shale)



Cooling & Solidification

METAMORPHIC ROCKS

- Formed from existing rocks under heat & pressure (e.g., Marble, Gneiss)



GEOMORPHOLOGY: Comprehensive Guide for UPSC Prelims 2026

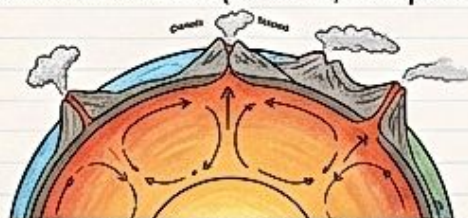
1 GEOMORPHIC PROCESSES: Forces shaping the Earth's Crust

ENDOGENIC FORCES (Internal Origin)

DIASTROPHISM

- Orogenic: Mountain building - fold, fault
- Epeirogenic: Continental building - uplift, subsidence)

SUDDEN MOVEMENTS (Volcanism, Earthquakes)

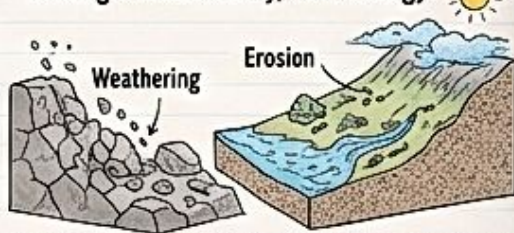


EXOGENIC FORCES (External Origin)

WEATHERING, MASS WASTING, EROSION, DEPOSITION

Agents: Water, Wind, Glaciers, Waves

Driving forces: Gravity, Solar Energy



2 THEORIES ON CONTINENTAL DRIFT & PLATE TECTONICS

CONTINENTAL DRIFT THEORY (Alfred Wegener, 1912)

Evidence:

- Jig-saw fit of continents
- Fossils (Glossopteris)
- Rock types
- Palaeoclimatic evidence (Tillite)

Supercontinent PANGAEA broke into LAURASIA & GONDWANALAND



SEA-FLOOR SPREADING (Harry Hess, 1960s)

Evidence:

- Mid-Oceanic Ridges (MORs)
- Age of rocks increasing away from MORs
- Magnetic anomalies (paleomagnetism)

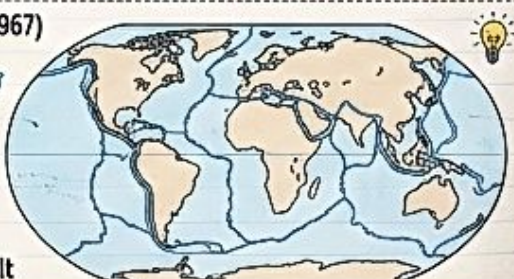


PLATE TECTONIC THEORY (McKenzie, Parker, Morgan, 1967)

- Earth's lithosphere divided into Tectonic Plates
- Driven by convection currents

Plate Boundaries:

DIVERGENT	CONVERGENT	TRANSFORM
Constructive	Destructive	Conservative
e.g. Mid-Atlantic Ridge	O-O, O-C, C-C, forming mountains/trenches	Conservative, Transform
		San Andreas Fault



3 EARTHQUAKES

CAUSES

- Release of stress along faults
- Volcanic activity
- Human activity (reservoirs)



TERMINOLOGY

- FOCUS (or Hypocenter) - origin point
- EPICENTER - point directly above on surface

WAVES

- Body Waves
 - P-waves: fastest through solids, liquids, gases
 - S-waves: slower, only through solids

& Surface Waves

- L-waves, R-waves: cause most damage
- SHADOW ZONES defined

DISTRIBUTION

- Circum-Pacific Belt ('Ring of Fire')
- Mid-Atlantic Ridge
- Alpine-Himalayan Belt



6 WEATHERING (In-situ disintegration)

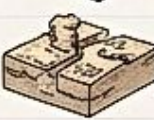
MECHANICAL

- Frost action
- pressure release (exfoliation), thermal stress



CHEMICAL

- Oxidation, carbonation, hydration, solution



BIOLOGICAL

- Plant roots, burrowing animals, human activity



4 VOLCANOES

SHIELD VOLCANOES

- Built from fluid basaltic
- Broad gentle slopes
- Non-explosive
- e.g. Mauna Loa



COMPOSITE VOLCANOES (STRATOVOLCANOES)

- Alternating layers of viscous lava, ash, and
- Steep conical shape
- Explosive
- e.g. Mt. Fuji



Distribution

- Convergent & Divergent boundaries
- Hotspots

5 TYPES OF MOUNTAINS

FOLD MOUNTAINS

Large scale folding of lithosphere e.g., Himalayas, Alps, Andes



BLOCK MOUNTAINS

Created by faulting and displacement of crust e.g., Sierra Nevada (Horst/Graben)



VOLCANIC MOUNTAINS

Build-up of erupted materials



EROSIONAL & DEPOSITIONAL LANDFORMS

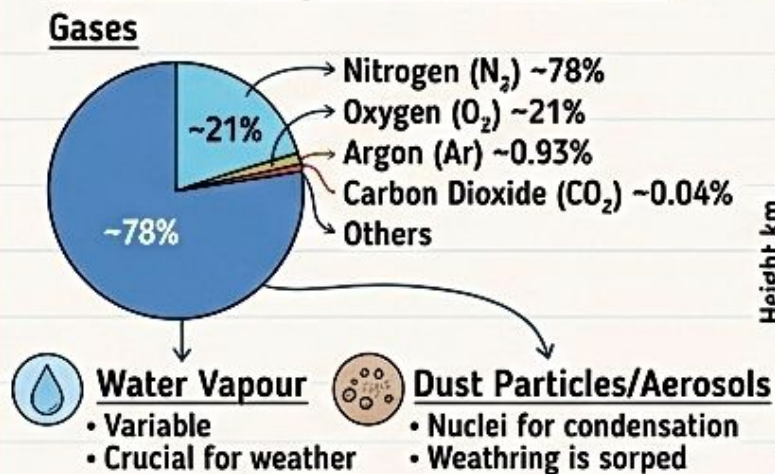
Agents:

	RIVER	WIND	GLACIER
E:	V-shaped valleys, Gorges, Waterfalls	Mushroom rocks, Yardangs, Deflation hollows	U-shaped valleys, Cirques, Aretes, Horns
D:	Meanders, Oxbow lakes, Floodplains, Deltas	Sand dunes, Loess	Moraines, Eskers, Drumlins

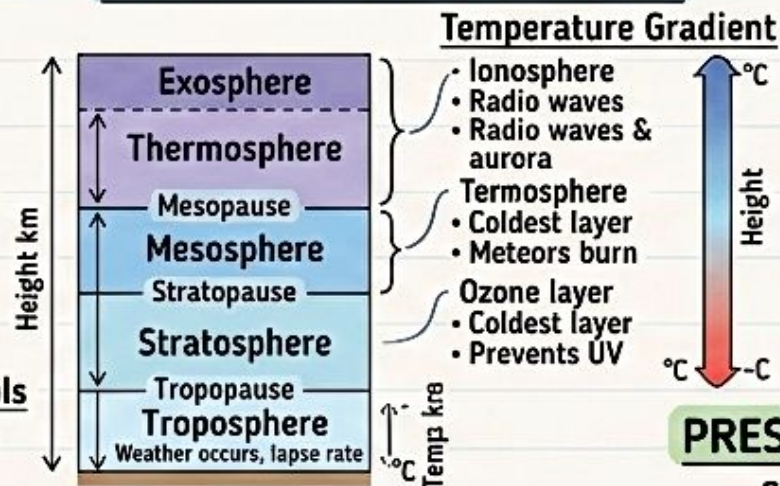


CLIMATOLOGY NOTES FOR UPSC 2026

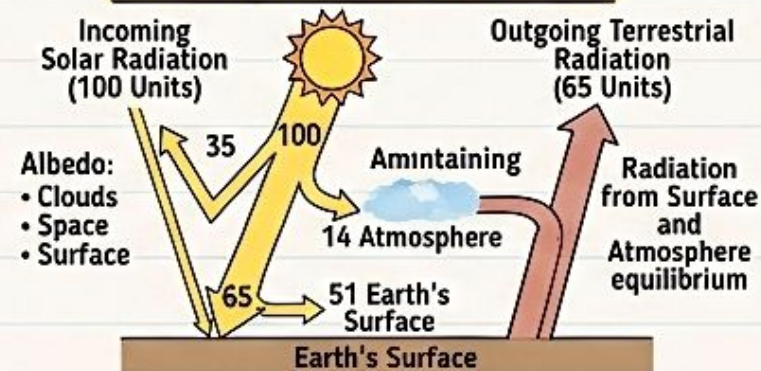
COMPOSITION OF ATMOSPHERE



STRUCTURE OF ATMOSPHERE

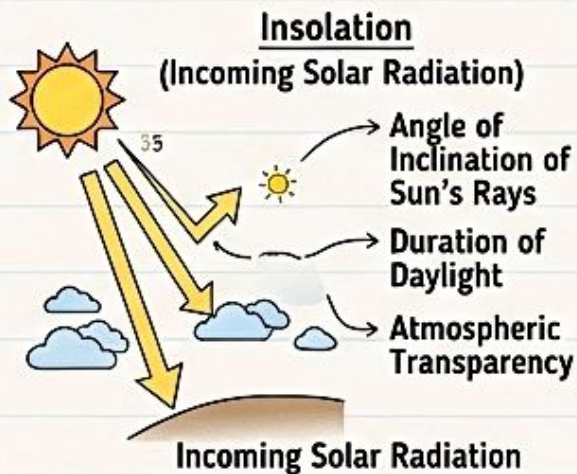


HEAT BUDGET OF EARTH

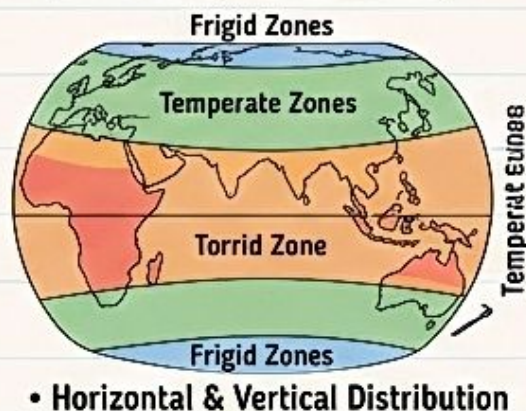


PRESSURE BELTS & WIND SYSTEMS

GLOBAL PRESSURE & WIND BELTS



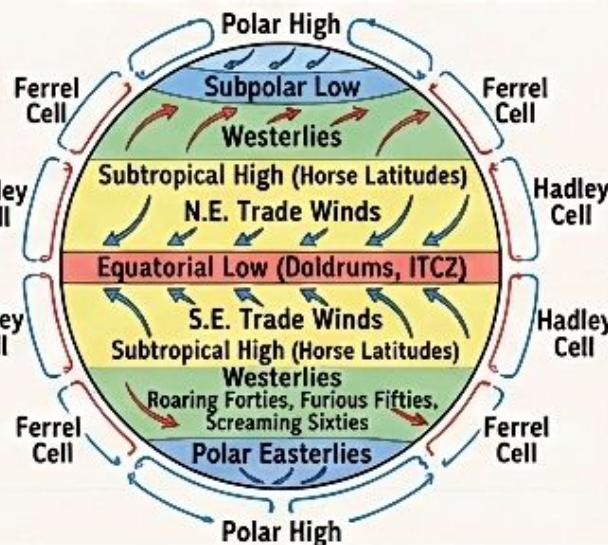
Global Temperature Zones



N. Hemisphere
S. Hemisphere

Coriolis Force

Left; S. Hemisphere



TRADE WINDS

- Steady, ocean routes
- Moisture for east coasts

PLANETARY WINDS

WESTERLIES

- Strong, variable weather
- Mid-latitudes

COMPREHENSIVE CLIMATOLOGY NOTES for UPSC PRELIMS 2026

SUBJECT: GEOGRAPHY

PREPARED BY: ABHISHA AI BY ABHISHEK PATHAK

1 INDIAN MONSOON MECHANISM



- Classical Theory: Diff. Heating (Land & Sea)
- SUMMER: Low Pressure over NW India/Tibet, High Pressure over Indian Ocean.
- Winds flow from Sea to Land (SW Monsoon).
- ITCZ moves North.



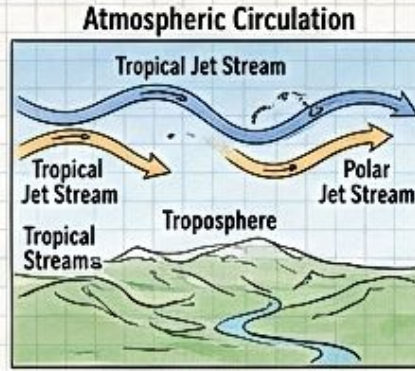
Summer Monsoon
(SW Flow, High PPT)



Winter Monsoon
(NE Flow, Dry, PPT in Tamil Nadu)

- Factors: Himalayas block cold air; Indian Ocean Dipole (Positive=Better Monsoon)

2 JET STREAMS & THEIR SIGNIFICANCE

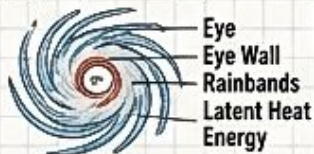


Atmospheric Circulation

- Definition: High-velocity, narrow upper-air westerlies (9-12 km alt.)
- MEANDERING PATHS: Rossby Waves
- TYPES:
 1. Polar Front Jet (Subpolar Lows)
 2. Subtropical Westerly Jet (Subtropical Highs).
- SIGNIFICANCE: Influence Weather Patterns; Affect Monsoon Onset (STWJ shifts North of Himalayas); Path for Cyclones; Aviation Efficiency.

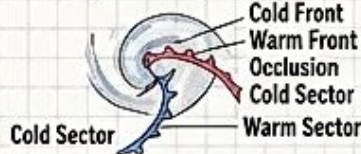
3 CYCLONES: TROPICAL VS. TEMPERATE

TROPICAL CYCLONE (e.g., Bay of Bengal)



Origin: Warm Tropical Oceans ($>27^{\circ}\text{C}$)
Size: Smaller (~200 km)
Movement: East to West (Trade Winds)
Energy: Latent Heat from Condensation
EYE: Clear, Low Pressure, Calm
Structure: Spiral Bands
IMPACT: Storm Surge, Coastal Damage

TEMPERATE CYCLONE (Extra-Tropical)



Origin: Mid-latitudes ($30-60^{\circ}$)
Size: Very Large (~1000 km)
Movement: West to East (Westerlies)
Energy: Temperature Contrast (Fronts)
EYE: No distinct clear eye
Structure: Comma-shaped (Frontal)
IMPACT: Widespread rain/snow over days

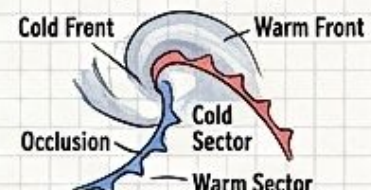
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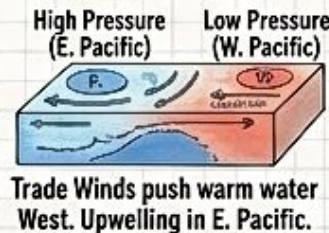
TEMPERATE CYCLONE (Extra-Tropical)



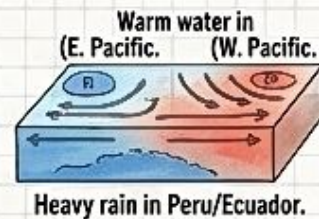
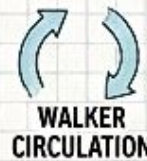
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4 EL NIÑO & LA NIÑA

NORMAL YEAR



Trade Winds push warm water West. Upwelling in E. Pacific.



EL NIÑO YEAR

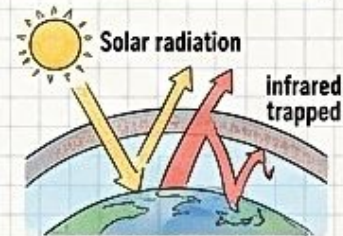
- Weakened Trade Winds.
- Warm water flows East.
- Drought in Australia/Asia, Heavy rain in Peru/Ecuador.
- Bad for Indian Monsoon.

LA NIÑA YEAR

- Intensified Trade Winds.
- Colder water in E. Pacific.
- Good for Indian Monsoon, Wet in Asia/Australia, Dry in America.

5 CLIMATE CHANGE & GLOBAL WARMING

- GLOBAL WARMING: Long-term rise in Earth's temperature due to enhanced Greenhouse Effect.
- Causes:
 - Burning Fossil Fuels (CO_2),
 - Deforestation,
 - Agriculture (Methane, N_2O),
 - Industrial Processes.
- IMPACTS:
 - Sea-level Rise (Ice Melt, Thermal Expansion)
 - Extreme Weather (Heatwaves, Droughts, Floods)
 - Ocean Acidification,
 - Biodiversity Loss,
 - Displacement.



GREENHOUSE EFFECT

- MITIGATION/ADAPTATION: Renewable Energy, Efficiency, Reforestation, Carbon Capture (CCS), Sustainable Practices, Policy.
- Key Pacts: Kyoto Protocol, Paris Agreement (Limit warming to 1.5°C)



OCEANOGRAPHY: OCEAN RELIEF & CURRENTS

– UPSC PRELIMS 2026 PREPARATION



OCEAN RELIEF: A CROSS-SECTION

Continental Shelf

- Gentle slope
- Photic zone
- Rich marine life
- <200m depth

Continental Slope

- Steep drop-off
- Boundary to deep ocean
- <2000m depth

Oceanic Ridge

Underwater mountain range, Plate boundary

Abyssal Plains

- Vast, flat deep ocean floor
- Sediment covered
- Unique organisms
- 3000-6000m depth

Oceanic Ridge

- Underwater mountain range
- Plate boundary
- Hydrothermal vents

Trench

- Deepest part
- Oceanic subduction e.g., Mariana Trench
- 11000m depth

Seamount / Guyot

Isolated underwater volcanic mountain

CAUSES OF CURRENTS

Wind

(Dominant driver)

Coriolis Force

(Deflects currents)

Temperature & Salinity differences
(Thermohaline circulation)

Coastline shapes
(Directs flow)

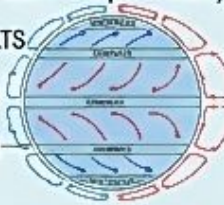
GYRES

(Major current patterns)

WIND BELTS

ISYDRLES

CORIOLIS



SEA MOUNT vs GUYOT
(Pointed vs Flat-topped)

OCEAN CURRENTS: GLOBAL CIRCULATION

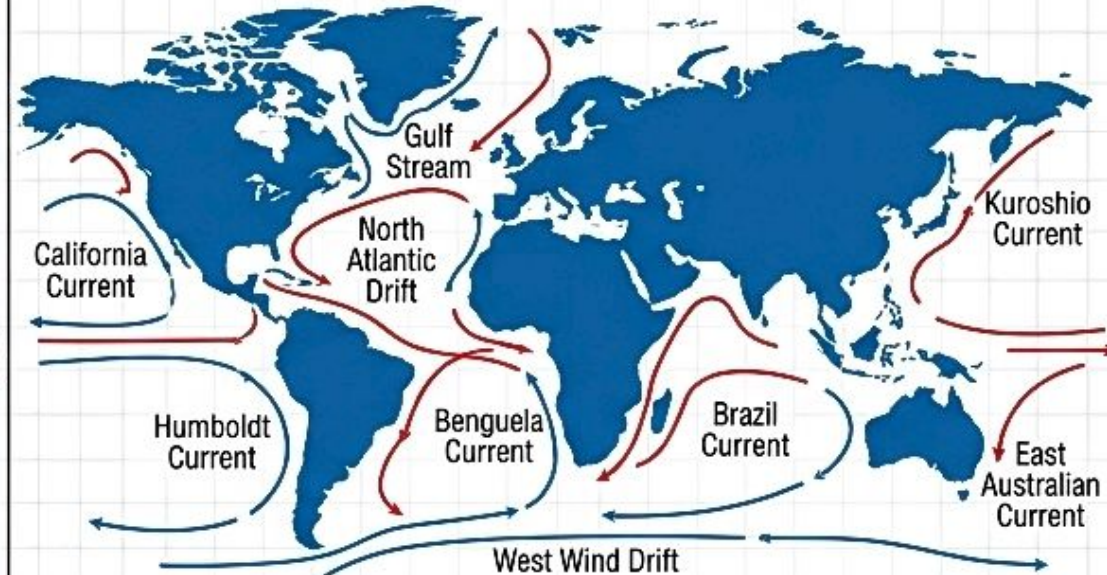
Cold Currents

Origin (Polar regions or deep ocean upwelling)

Characteristics (Lower temperature & salinity, flow equatorward)

Impacts (Dry/Desert conditions on west coasts, e.g., Atacama)

GYRES (Major current patterns)



Cold Currents

Origin (Polar regions or deep ocean upwelling)

Characteristics (Lower temperature

Impacts (Dry/Desert conditions on west coasts, e.g., Atacama)

Warm Currents

Origin (Tropics)

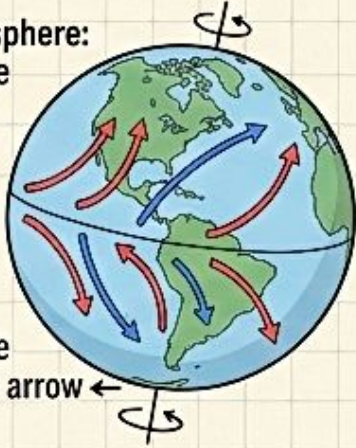
Characteristics (Higher temperature & salinity, flow poleward)

Impacts (Warm coastal climates, create fog, e.g., North Atlantic)

1. CORIOLIS EFFECT: Apparent Deflection

ABHISHA AI

Northern Hemisphere:
Deflection to the
the Right; path
curve arrow →

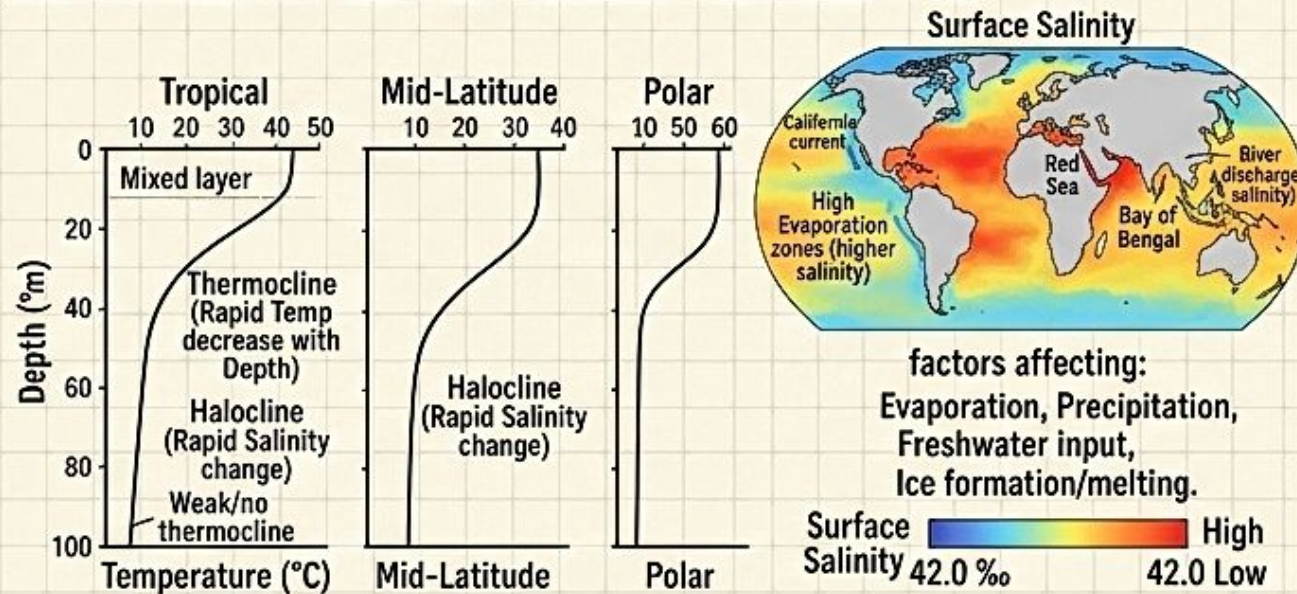


Southern Hemisphere:
Deflection to the
Left; path curve arrow ←

- Causes: Earth's Rotation
- Latitude dependence: Zero at Equator, Maximum at Poles
- Velocity dependence: Increases with object speed
- Crucial for Trade Winds, Westerlies, and Ocean Gyres.

ABHISHA AI

3. SALINITY & TEMPERATURE DISTRIBUTION



2. TIDES: Gravitational Dynamics

ABHISHA AI

SPRING TIDES



Highest High Tides; Lowest Low Tides.

Maximum Tidal Range.

NEAP TIDES

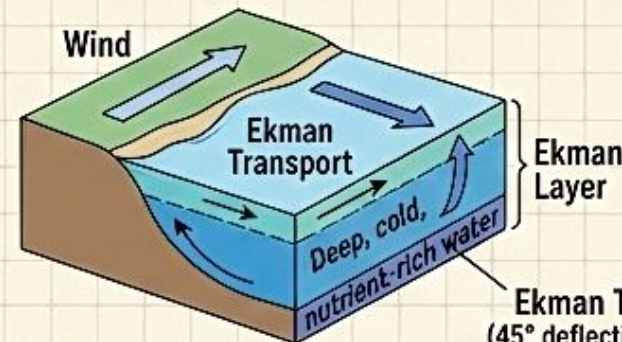


Lowest High Tides; Highest Low Tides.

Minimum Tidal Range.

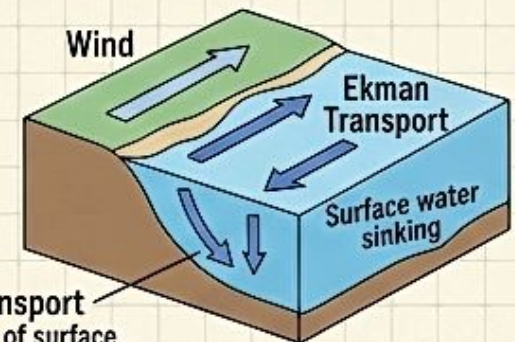
4. UPWELLING & DOWNWELLING: Global Circulation

COASTAL UPWELLING



Nutrient-rich zone; supports highly productive marine ecosystems and fisheries.

COASTAL DOWNWELLING



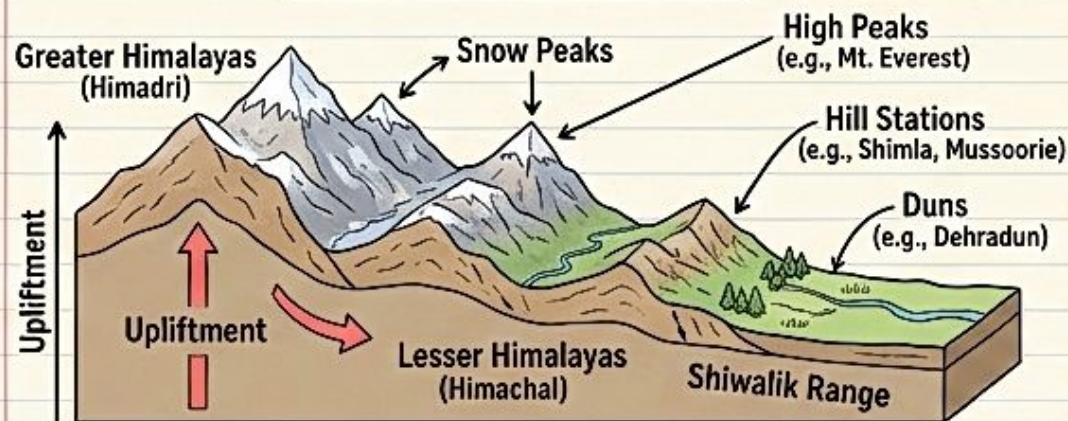
Ekman Transport (45° deflection of surface currents to the wind direction; overall transport 90°)

Nutrient-poor surface water moves downward.

COMPREHENSIVE NOTEBOOK: PHYSIOGRAPHY OF INDIA FOR UPSC PRELIMS 2026



HIMALAYAN DIVISIONS



NORTHERN PLAINS FORMATION



Indus → Lugal → Milar → Tamgas → Tapi

Time

Himalayan Rivers
(Indus, Ganga, Brahmaputra)

Depression
(Foredeep)

Deposition
of Alluvium

Aggradational Plain

Bhabar (Porous, pebbles)

Terai (Marshy, thick forest)

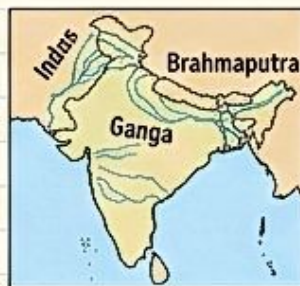
Bhangar (Older alluvium, Kankar)

Khadar (Newer alluvium, fertile)



Himalayan Rivers
(Indus, Ganga, Brahmaputra)

Deposition
of Alluvium



PENINSULAR PLATEAU



Features:

- Hard Crystalline Rocks
- Gentle Slopes
- Major Rivers Flow East (Except Narmada, Tapi)



W. Ghats:

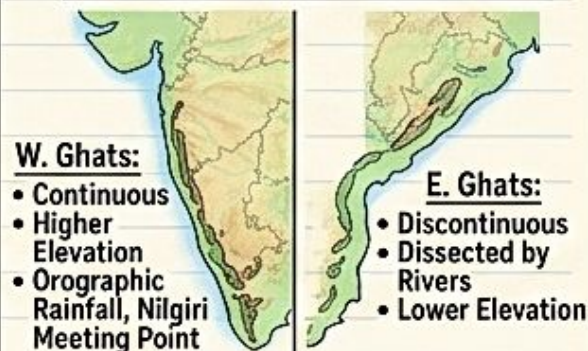
- Continuous
- Higher Elevation
- Orographic Rainfall
- Nilgiri Meeting Point

E. Ghats:

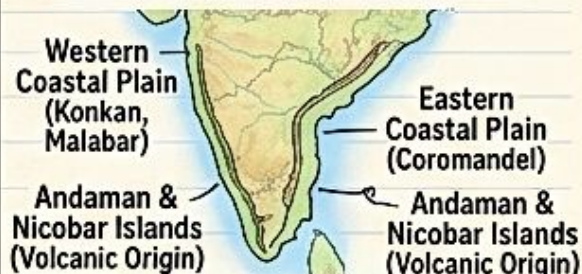
- Discontinuous
- Dissected by Rivers
- Lower Elevation



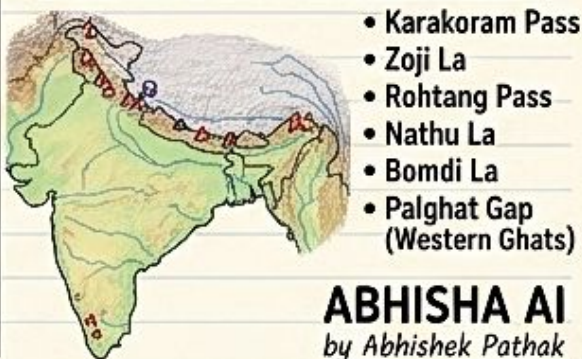
WESTERN vs EASTERN GHATS



COASTAL PLAINS & ISLANDS



IMPORTANT MOUNTAIN PASSES



ABHISHA AI
by Abhishek Pathak



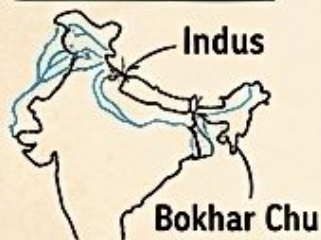
INDIAN DRAINAGE SYSTEM - COMPREHENSIVE NOTES FOR UPSC PRELIMS 2026



ABHISHA AI by Abhishek Pathak

1 HIMALAYAN RIVERS

INDUS SYSTEM

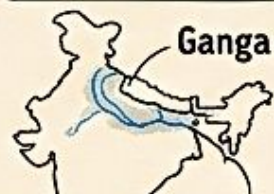


• Key points

↳ TRIBUTARIES

- Jhelum • Chenab
- Ravi • Beas
- Beas • Sutlej

GANGA SYSTEM

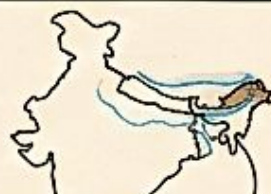


• Key point: Devprayag/Alaknanda+Bhagirathi

↳ TRIBUTARIES

- Yamuna • Gandak
- Ghaghara • Kosi
- Gandak

BRAHMAPUTRA SYSTEM



• Origin: Chemayungdung

↳ Key points

- Dibang • Lohit
- Lohit • Subansiri
- Manas • Teesta

2 PENINSULAR RIVERS

GODAVARI ("DAKSHIN GANGA")



• Key Points: Penganga

• Key point; Aldravati

↳ TRIBUTARIES

- Penganga
- Indravati
- Pranhita
- Manjira

KRISHNA



↳ TRIBUTARIES

- Tungabhadra
- Bhima
- Musi

NARMADA



• Flows in Rift Valley

• Amarkantak origin

WEST-FLOWING RIVERS (No Deltas, Estuaries)

- Narmada
- Tapi

3 RIVER LINKING PROJECTS (NRLP)



KEN-BETWA LINK

Phase II Underway (2026 Budget)

↳ Submergence in Panna TR

Other Major Links

→ Damanganga-Pinjal

→ Godavari-Krishna



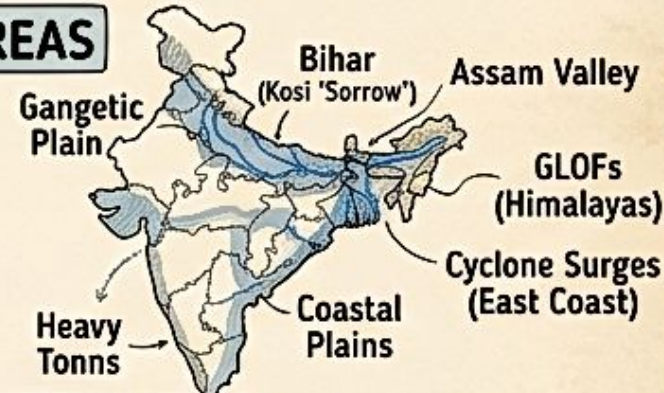
4 FLOOD-PRONE AREAS

Causes of Brannosolors:

- ↳ Gangetic Plain
- ↳ Brahmaputra Valley
- ↳ parts of Coastal Plains

Causes:

- Heavy rain
- Silt • Encroachment



5 MAJOR DAMS & RESERVOIRS



TEHRI DAM	Bhagirathi River	Uttarakhand	Highest Dam
HIRAKUD DAM	Mahanadi River	Odisha	Longest Earthen Dam
BHAKRA NANGAL	Sutlej River	HP/Punjab	Gobind Sagar Reservoir
SARDAR SAROVAR	Narmada River	Gujarat	Multipurpose
SRISAILAM DAM	Krishna River	AP/Telangana	
METTUR DAM	Kaveri River	Tamil Nadu	



INDIAN CLIMATE & MONSOON: HAND-DRAWN NOTES BY ABHISHA AI (BY ABHISHEK PATHAK)

ABHISHA AI

1 FACTORS AFFECTING INDIAN CLIMATE



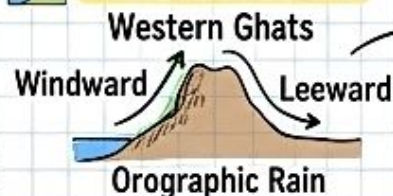
LATITUDE



Tropic of Cancer



PHYSIOGRAPHY



THE HIMALAYAS



Orographic Rain

- Savor hality: sin &
- Interior rain

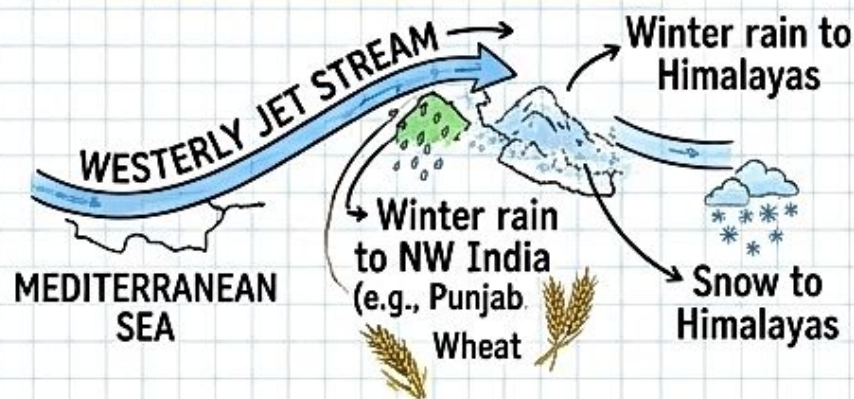
DISTANCE FROM THE SEA

- Continenality
- Coastal moderation
- Interior extremes

JET STREAMS



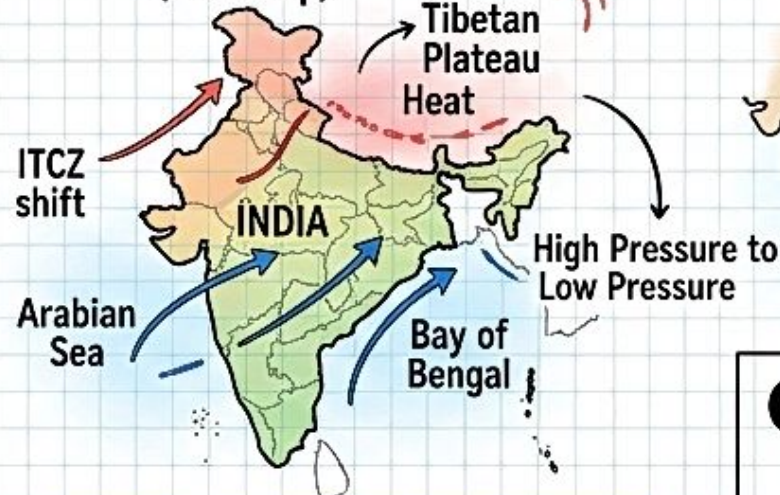
4 WESTERN DISTURBANCES



2 THE MONSOON SYSTEMS

SW MONSOON

(June-Sep)



NE MONSOON

(Oct-Dec)



3 RETREATING MONSOON

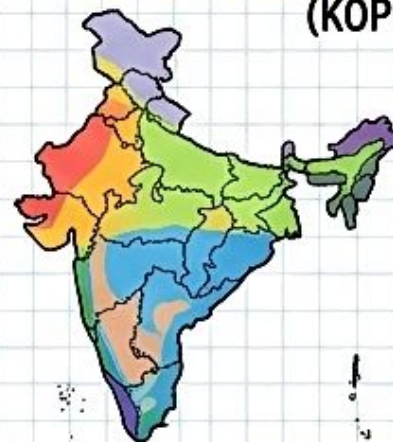
- October Heat
- Transition period
- Decreasing rain
- High humidity & temp

5 CYCLONE-PRONE REGIONS

HIGH RISK ZONES



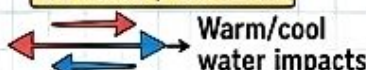
6 CLIMATIC REGIONS OF INDIA (KOPPEN'S)



- Amw: Monsoon short dry (West Coast)
- Aw: Tropical Savanna (Peninsular Plateau)
- As: Monsoon dry summer (Coromandel)
- Cwg: Humid Sub-tropical (Ganga Plain)
- BShw: Semi-arid Steppe (Rain-shadow)
- BWhw: Hot Desert (W. Rajasthan)
- E: Polar (Himalayas)

IMPORTANT NOTES

EL NIÑO / LA NIÑA

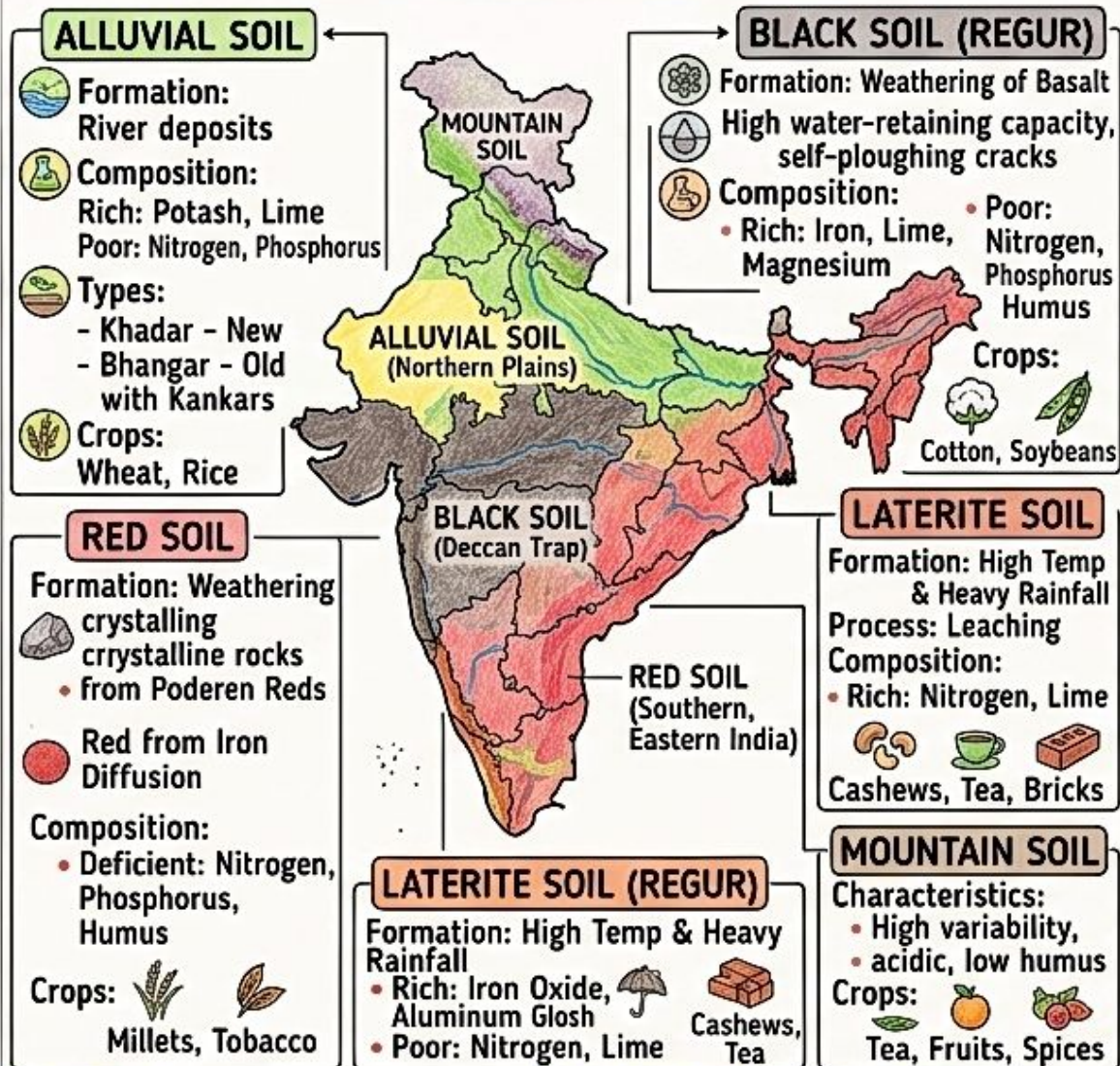


INDIAN OCEAN DIPOLE (IOD)

- Positive IOD good for Monsoon

SOILS OF INDIA & CONSERVATION (UPSC PRELIMS 2026)

MAJOR SOIL TYPES OF INDIA



Pro-tip: Most Indian soils are deficient in Nitrogen & Humus

SOIL EROSION & CONSERVATION METHODS

CAUSES OF SOIL EROSION

Water Erosion



Sheet



Gully/Ravines like Chambal

Wind Erosion



Arid areas

Anthropogenic



Overgrazing, Deforestation, Faulty farming

CONSERVATION METHODS

Contour Ploughing



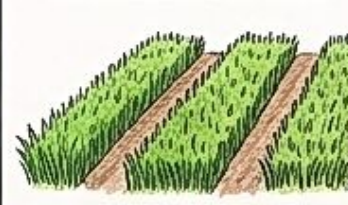
Ploughing along contours to reduce flow

Terrace Farming



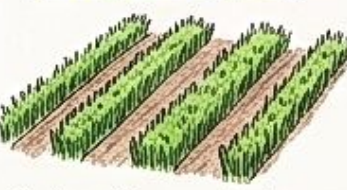
Steps cut into slopes, Himalayas

Strip Cropping



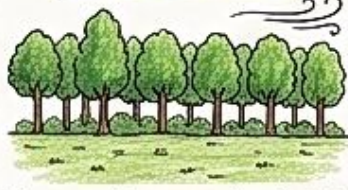
Strips of grass between crops on Rajasthan

Strip Cropping



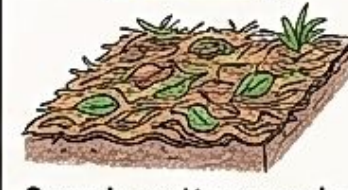
Strips of grass between crops on Rajasthan

Shelter Belts



Rows of trees planted to block wind, Rajasthan

Mulching



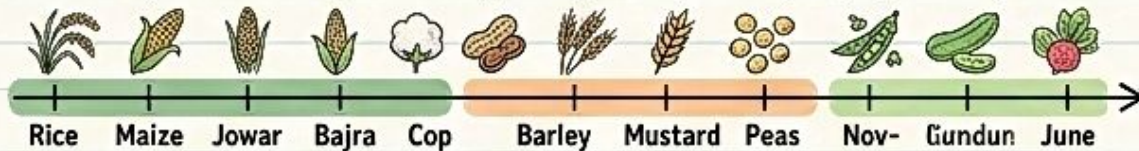
Organic matter covering ground to retain moisture

ABHISHA AI by Abhishek Pathak

UPSC PRELIMS 2026: AGRI-NOTEBOOK - COMPREHENSIVE GUIDE

I. CROPPING SEASONS

- KHARIF** (SW Monsoon, June-Oct): Rice, Maize, Jowar, Bajra, Cotton, Groundnut



- RABI** (Winter, Nov-April): Wheat, Barley, Gram, Mustard, Peas

- ZAID** (Summer, Mar-June): Watermelon, Cucumber, Fodder Crops

II. GREEN REVOLUTION

- ORIGIN** (1960s): Dr. M.S. Swaminathan, Norman Borlaug
- FEATURES**: HYV Seeds, Chemical Fertilizers, Irrigation
- FOCUS**: Wheat & Rice
- SUCCESS**: Seed-Water-Fertilizer Package
- IMPACT**: Food Security, regional disparities
- GR 2.0 (RAINBOW)**: Diversification, Eastern States, PDS, Soil Health, Sustainable Farming.

III. MAJOR CROPS & PRODUCING STATES

RICE	WB, UP, PUNJAB, AP
WHEAT	UP, PUNJAB, HARYANA, MP
PULSES	MP, MAHARASHTRA, RAJASTHAN
OILSEEDS	GUJARAT, RAJASTHAN, MP
SUGARCANE	UP, MAHARASHTRA, KARNATAKA
COTTON	GUJARAT, MAHARASHTRA, TELANGANA
MILLET (Shree Anna)	RAJASTHAN, KARNATAKA, MAHARASHTRA



IV. MSP & AGRICULTURAL REFORMS

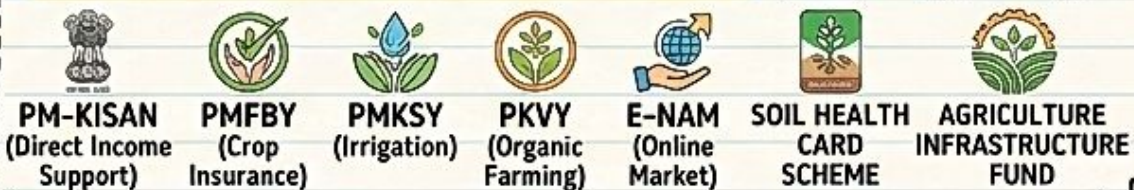
- MSP** (Minimum Support Price): Announced by Govt (CCEA) based on CACP, fixed at least 1.5x Production Cost (A2+FL). Covers 22+1 crops.
- REFORMS**: APMC (Agricultural Produce Market Committees), eNAM (Electronic National Agricultural Market), Contract Farming, FPOs (Farmer Agricultural Market), Contract Farming, FPOs (Farmer Producer Organisations) & Agri-Startups, land Leasing, Supply Chain Mgmt.

V. IRRIGATION METHODS

- WELLS & TUBEWELLS** (Largest share)
- WELLS & IRRIGATION** (Largest share)
- CANAL IRRIGATION** (Surface water, large projects)
- TANK IRRIGATION** (Peninsular India)
- MICRO-IRRIGATION**:

- DRIP IRRIGATION** (Efficient, precise, for orchard crops),
- SPRINKLER IRRIGATION** (Overhead, suitable for sandy soil). Less water waste.

VI. AGRICULTURAL SCHEMES



INDIA'S MINERAL AND ENERGY RESOURCES: A NOTEBOOK GUIDE FOR UPSC PRELIMS 2026

COMPILED BY: ABHISHA AI | Abhishek Pathak

COAL DISTRIBUTION (GONDWANA & TERTIARY)

GONDWANA COAL (98%)

- Jharia (Jharkhand - largest)
- Raniganj (West Bengal)
- Bokaro (Jharkhand)
- Korba (Chhattisgarh)
- Singrauli (MP/UP)
- Talcher (Odisha)

TERTIARY COAL (LIGNITE)

- Neyveli (Tamil Nadu - largest)
- Palana (Rajasthan)
- Lakhimpur (Assam)

PETROLEUM & NATURAL GAS REGIONS

MAJOR REGIONS

- Mumbai High (offshore - largest)
- Assam (oldest; Digboi, Naharkatiya, Moran-Hugrijan)
- Gujarat (Ankleshwar, Kalol, Mehsana)
- KG Basin (Krishna-Godavari, natural gas - Reliance D6 block)

METALLIC MINERALS: IRON ORE, BAUXITE, MANGANESE

IRON ORE BELTS

- Odisha-Jharkhand (Badampahar, Kiruburu, Noamundi)
- Durg-Bastar-Chandrapur (Bailadila mines - high grade export)
- Bellary-Chitradurga-Chikkamagaluru Kudremukh mines - slurry transport

BAUXITE (ALUMINIUM)

- Odisha (largest producer)
- Panchpatmali (Koraput)
- Amarkantak plateau (MP)
- Maikal hills

MANGANESE (STEEL MAKING)

- Balaghat (MP)
- Nagpur-Bhandara (Maharashtra)
- Kendujhar (Odisha)

ATOMIC MINERALS & RENEWABLE POTENTIAL

ATOMIC MINERALS

- Uranium (Jaduguda, Tummalapalle - Andhra Pradesh)
- Thorium (Monazite sands in Kerala, Odisha)
- Lithium (Recent: Reasi J&K, Mandya Karnataka)

RENEWABLE POTENTIAL

- Solar (748 GW; Bhadla Solar Park Rajasthan, Gujarat lead)
- Wind (302 GW; Tamil Nadu, Gujarat, Maharashtra lead)

★ KEY FOCUS FOR UPSC: Spatial distribution, major mines, and geological belts

UPSC PRELIMS 2026: INDIAN TRANSPORT & COMMUNICATION (NOTEBOOK SUMMARY)



MAJOR PORTS (13)

WEST COAST

- Kandla (Deendayal)
- Mumbai
- JNPT (Nhava Sheva)
- Mormugao
- New Mangalore
- Kochi

EAST COAST

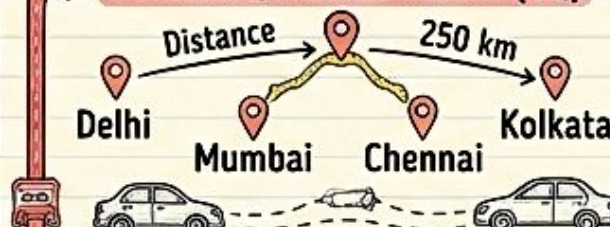
- Tuticorin
- Chennai
- Ennore (Kamarajar)
- Visakhapatnam
- Paradip
- Haldia
- Kolkata (SP Mukherjee)
- Natural (e.g., Mumbai, Vizag)
- Artificial (e.g., Chennai, JNPT)

→ upcoming VadHAVAN Port (green stim)



NATIONAL HIGHWAYS (NH)

GOLDEN QUADRILATERAL (GQ)



NS-EW CORRIDOR

- Srinagar to Kanyakumari (NH 44 - Longest)
- Porbandar to Silchar (NH 27)

Intersection point: Jhansi

- GATI SHAKTI
- BHARATMALA PRIYOJANA
- Traffic lights and bridges



INLAND WATERWAYS (NW)



Major NWs	ROUTE	RIVER	length
NW-1	Prayagraj-Haldia	Ganga	→→→→
NW-2	Prasagraj-Prayagraj-Haldia	Ganga-Hooghly	→→→→
NW-3	Prayagraj-Haldia	Ganga	→→→→
NW-4	Prayagraj-Haldia	Ganga	→→→→
NW-5	Prayagraj-Haldia	Ganga	→→→→

→ "SAGARMALA PROJECT": synergy for times of by India's ports



RAILWAY ZONES (19)

MAJOR ZONES HEADQUARTERS

- Northern New Delhi
- Southern Chennai
- Central Mumbai CSMT
- Western Mumbai WR
- Southern Mumbai WR
- Mournern Visakhapatnam

★ Newest zone : South Coast Railway (SCoR) - Visakhapatnam

★ KONKAN RAILWAY

- Railway + steam train
- Plothan hi milhk away

★ DIAMOND QUADRILATERAL (HSR)

- Steam zones: stdness
- Diesel (coombapteran)
- Modern train modern

INTERNATIONAL TRADE ROUTES & CONNECTIVITY



→ INSTC (International North-South Transport Corridor)

Route: Mumbai, Iran, Russia

Advantages:

- Mumbai, Beaca & Advaveraon
- Increases Natik east analysis



→ CHABAHAIR PORT (Iran) & its role:

- Connector, Iran and
- Connection: Modair & CHABAHAIR PORT



→ IMEC (India-Middle East-Europe Economic Corridor)

- Kolta-Sittwe-Mizoram
- Arrow lines
- Containerst & cargo sines



→ KALADAN Multi-Modal Transit Transport Project (Kolkata-Sittwe-Mizoram)



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by Abhishek Pathak

UPSC PRELIMS 2026: TRANSPORT & COMMUNICATION IN INDIA

MAJOR PORTS (13) & NEW ONE



NATIONAL HIGHWAYS (NH)

Golden Quadrilateral (GQ)



N-S & E-W Corridors

Crossing point Jhansi

→ Connectivity → Economic Growth

- ◆ Bharatmala Pariyojana
- ◆ 2% Roads → 40% Traffic
- ◆ Logistics improvement

INLAND WATERWAYS (NW)

Main NWs

- NW-1: Prayagraj to Haldia (Ganga, 1620 km) **Longest**
- NW-2: Sadiya to Dhubri (Brahmaputra, 891 km)
- NW-3: West Coast Canal (Kerala, 205 km)
- NW-4: Kakinada-Puducherry

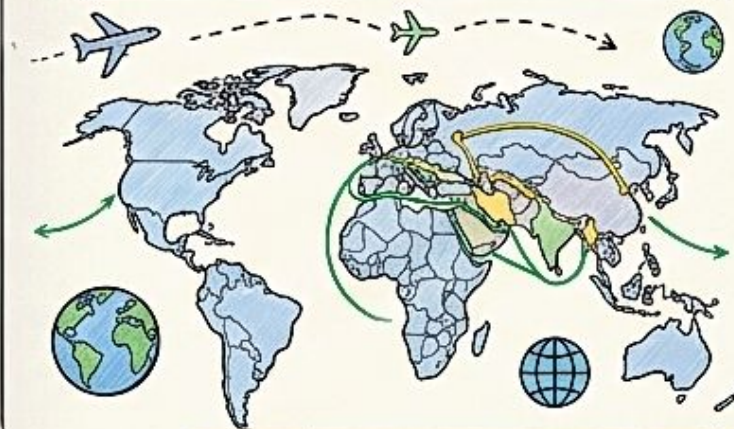
- ◆ National Waterways Act 2016 (111 NWs)
- ◆ Cheaper, Eco-friendly transport
- ※ Jal Marg Vikas Project (World Bank)

RAILWAY ZONES & HEADQUARTERS

19 ZONES

- NORTH:** New Delhi (NDLS)
 - SOUTH:** Chennai (MAS)
 - EAST:** Kolkata (HWH)
 - WEST:** Mumbai Churchgate (CCG)
 - CENTRAL:** Mumbai CSMT
 - SOUTH COAST (New):** Visakhapatnam (VSKP)
- Metro Railway Zones:**
- Kolkata
 - NDLS
 - ABP

INTERNATIONAL TRADE ROUTES



INSTC (International North-South)

- Mumbai to Moscow via Iran (Chabahar) and Azerbaijan
- Chabahar Port → Gateway to Afghanistan, bypassing Pakistan

IMEC (India-Middle East-Europe)

- India to Europe via ship-to-rail (UAE, Saudi, Jordan, Israel)

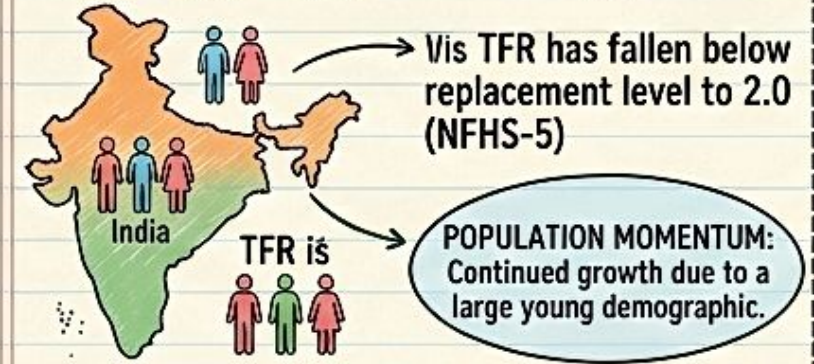
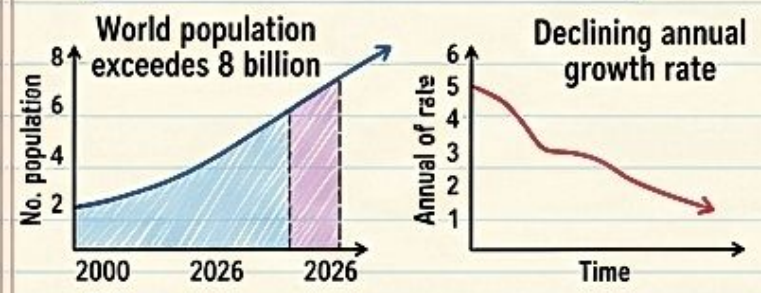
India-Myanmar-Thailand Highway

- Moreh (Manipur) to Mae Sot
- Key Act East
- ◆ Strategic ties, energy security
- ◆ Bypassing land blocks

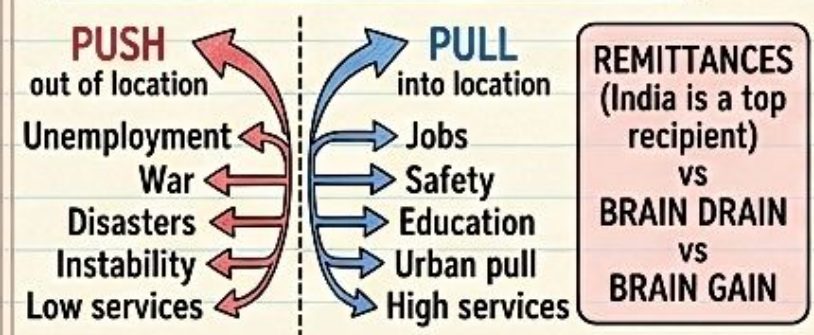
POPULATION GEOGRAPHY

COMPREHENSIVE NOTES (FOR PRELIMS)

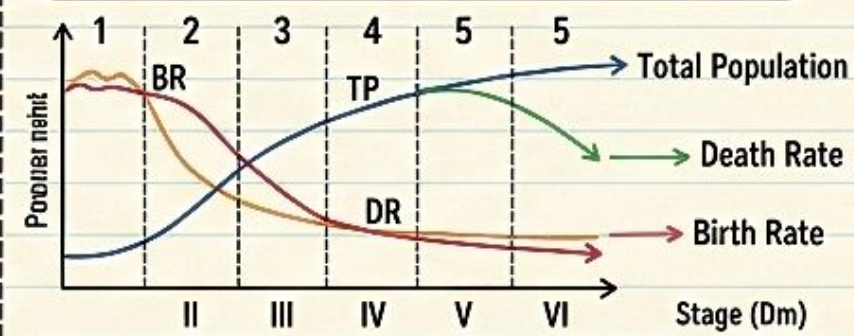
POPULATION GROWTH TRENDS



MIGRATION: PUSH & PULL FACTORS

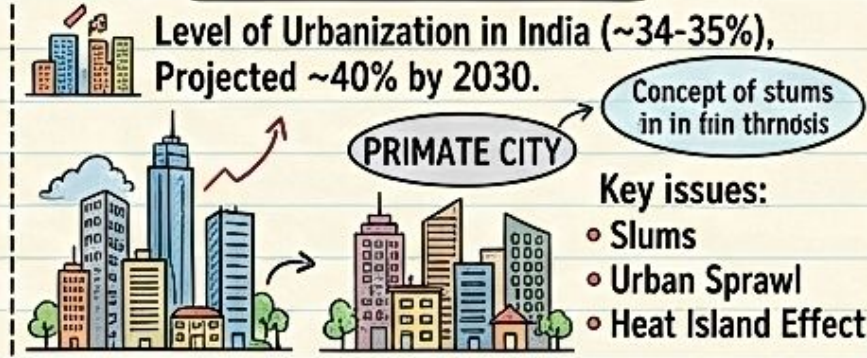


DEMOGRAPHIC TRANSITION MODEL (DTM)

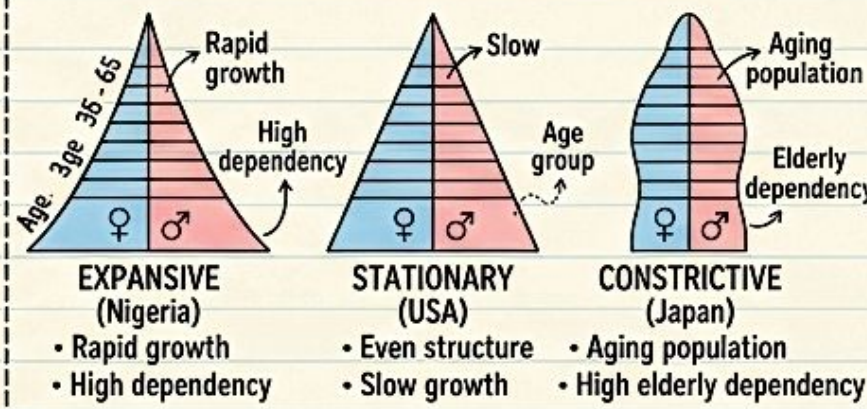


Stage	Characteristics	Examples
Stage 1	Pre-industrial	
Stage 2	Rapid Growth (African nations)	Pre-industrial
Stage 3	Slowing Growth (India)	India
Stage 4	Low/Stable (USA/China)	USA/China
Stage 5	Negative/Declining (Japan)	Japan

URBANIZATION TRENDS



POPULATION PYRAMID TYPES



CENSUS DATA INTERPRETATION (FOCUS 2026)

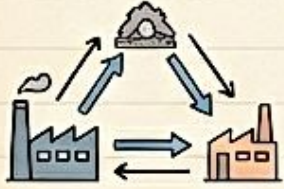
- India's First DIGITAL CENSUS
- Self-Enumeration option
- LITERACY RATE : (7+ aged, can read/write)
- SEX RATIO : (Females/1000 Males)
- DEPENDENCY RATIO : (Non-working 0-14 & 65+ / Working-age 15-64)

STRATEGY TIP > Analyze spatial patterns in Indian states for marriage for TFR and urbanization (Bihar vs Kerala, Goa/Tamil Nadu)

NOTEBOOK STUDY GUIDE: ECONOMIC GEOGRAPHY (FOR UPSC PRELIMS 2026)

1. INDUSTRIAL LOCATION FACTORS

- Minimizing Costs (Transport, Labor, Energy)
- Weber's Theory: Weight-Losing Industries (e.g., Iron & Steel - located near Raw Materials)



- Footloose Industries (e.g., IT, Electronics - located anywhere with infrastructure connectivity & skilled labor)

2. TYPES OF INDUSTRIES



Mineral-based

- Iron & Steel
- Cement



Agro-based

- Cotton
- Sugar
- Food Processing



Forest-based

- Paper
- Lac



Sunrise Industries

- IT, Wellness
- Biotech
- high growth

3. SEZs AND INDUSTRIAL CORRIDORS

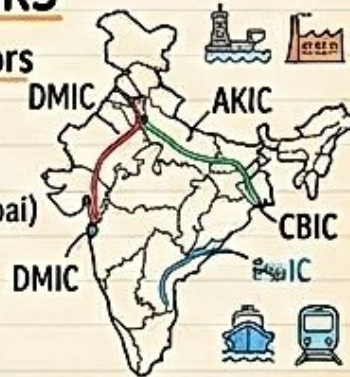
SEZs (Special Economic Zones)

- Duty-free zones
- focus on exports
- FDI attraction



Industrial Corridors

- Integrated infrastructure
- DMIC (Delhi-Mumbai) - Japan support, multi-modal



4. GLOBAL TRADE ROUTES



- Critical chokepoints → major significance due to ice melt.

Note - Arctic Routes: Gaining significance due to ice melt.



5. WORLD AGRICULTURE TYPES



- Intensive Subsistence (Rice - SE Asia)
- Commercial Grain Cultivation (Wheat, Corn - Prairies, Steppes)
- Plantation (Tea, Coffee - India, Brazil)
- Mediterranean Agriculture (Fruits, Wine - Southern Europe/Calif)
- Pastoral Nomadism (Central Asia, Africa)



6. RESOURCE DISTRIBUTION

Major deposits:

- Iron Ore (Pilbara-AUS, Carajas-Brazil)
- Coal (Shanxi-China, Appalachian-USA)

Critical Minerals:

- Lithium Triangle (Chile, Argentina, Bolivia)
- + Cobalt (DRC) - essential for batteries & EVs



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